MILITARY SPECIFICATION

RADIO SET AN/ARC-38A

This amendment forms a part of Military Specification MIL-R-22604(WEP) dated 15 August 1960 and has been approved by the Bureau of Naval Weapons, Department of the Navy.

*Page 1, paragraph 2.1, after "to the extent specified herein:": Add:

"(See 6.3)".

*Page 5, paragraph 3.3.1.1.2, after "signal-plus-noise-to-noise ratio of at least 6 db": Add:

"from 2.0000 mc to 14.2500 mc and at least 5 db from 14.2500 mc to 25.0000 mc."

*Page 6, paragraph 3.3.1.1.5: Delete and substitute:

"3.3.1.1.5 Selectivity, SSB/FSK - The selectivity of the receiver during the SSB/FSK mode of operation shall be as follows:

Attenuation below Channel Frequency Plus 1.0 kc/s	Frequency Deviation from Channel Frequency Plus 1.0 kc/s
6 db	-0.7 and +2.3 kc/s min.
60 db	-3.5 and +5.5 kc/s max."

*Page 6, paragraph 3.3.1.1.6.1: Delete and substitute:

"3.3.1.1.6.1 Internal Spurious Signals - The audio output which results from interaction of the various conversion oscillators and which is independent of a received signal shall not exceed 2.5 milliwatts. The sensitivity and volume controls shall be at the settings used during the measurement of sensitivity."

*Page 7, paragraph 3.3.1.1.11.2(1), Overall Fidelity: The first sentence shall be changed as follows:

"The overall audio response shall not vary more than 5 db for modulation frequencies of 300 to 3000 cps; the allowable 5 db variation shall be measured between the maximum and minimum response levels."

Page 9, paragraphs 3.3.1.2.9 through 3.3.1.2.11: Delete and substitute:

"3.3.1.2.9 AME Modulation -

3.3.1.2.9.1 Capability - When a 1000 cps voltage from the dummy microphone (3.2.15.2) of 1.4 peak open-circuit volts is applied, the transmitter RF output envelope shall be modulated by positive and negative amounts (equal within 5%) to at least 85%. The transmitter shall deliver rated P.E.P. output in accordance with 3.3.1.2.5 with the following 250-254 kc high and low level clipper (3.4.1.3.2) and threshold capabilities:

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(1) With the low level threshold adjustment "off", for the above 1.4 peak open circuit volts 1000 cps input, the high level clipper shall have a minimum (adjustable) capability of 15 db clipping as evidenced by maintaining the minimum 75% modulation for a 15 db reduction in 1000 cps input voltage. This capability can be determined either by measurement of the overall transmitter output or by equivalent measurement of the 250-254 kc output of the modulator module.

(2) The low level threshold circuit shall have a minimum (adjustable) capability such that a 0.7 peak open-circuit volt 1000 cps input can be made to produce a transmitter output envelope modulation of 10% (or less); the other adjustments being the same as determined for the capability defined in 3.3.1.2.9.1(1) above.

3.3.1.2.9.2 Fidelity - For modulation frequencies of 350 to 3200 cps, the carrier shall be modulated not less than 40% nor more than 70% when an audio frequency of 0.7 peak open-circuit volt is applied to the input circuit. Attenuation below 300 cps shall be as rapid as practicable. At 5000 cps the response shall be attenuated between 0 to 10 db from the 1000 cps level and at 6500 cps the response shall be attenuated at least 15 db from the 1000 cps level. The total harmonic content of the modulation, with input frequencies to the dummy microphone of 300 to 3000 cps under the above conditions, shall not exceed 18% as determined by audio distortion measurement of the envelope detected output. The over-all noise shall be at least 35 db below 95% modulation. This capability can be determined either by measurement of the overall transmitter output or by equivalent measurement of the 250-254 kc output of the modulator module. The low level clipping and threshold functions shall be disabled and a maximum of 6 db high level clipping shall be used when making the above measurements.

3.3.1.2.10 SSB/FSK Modulation -

3.3.1.2.10.1 Capability - When either a two tone signal input (equal amplitudes of 1000 cps and 1700 cps) or a single tone input (teletype signal frequencies of 1575 cps or 2425 cps) of 1.4 peak open circuit volts from the dummy microphone (3.2.15.2) is applied, the transmitter shall deliver rated P.E.P. output in accordance with 3.3.1.2.5 with the following 250-254 kc high and low level clipper (3.4.1.3.2) and threshold capabilities:

(1) With the low level threshold adjustment "off", for the above 1.4 peak open circuit volts two tone signal input, the high level clipper shall have a minimum (adjustable) capability of 15 db P.E.V. clipping. This capability can be determined either in the manner of 3.3.1.2.9.1(1) or by oscilloscope or equivalent measurement of the 250-254 kc P.E.V. at the output of the clipper-filter. In the latter case, the observed P.E.V. shall remain within 25% when the input audio voltage is reduced 15 db.

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(2) The low level threshold circuit shall have a minimum (adjustable) capability such that a 0.7 peak open circuit volt two tone signal input can be attenuated 18 db below, or to 12% of, the P.E.V. measured at the 250-254 kc output filter and with other adjustments the same as for 3.3.1.2.10.1(1) above. As an alternate, this capability can be measured as described in 3.3.1.2.9.1(2).

3.3.1.2.10.2 Linearity - With the two tone signal input level adjusted to produce rated power output in accordance with 3.3.1.2.5 and with the clipping and threshold functions disabled, all spurious rf output within the desired passband shall be attenuated at least 25 db below P.E.P. output level.

3.3.1.2.11 Automatic Level Control - The automatic level control circuit shall operate to provide rated power output as specified in 3.3.1.2.5 and to maintain the linearity and spurious output performance specified in 3.3.1.2.7 and 3.3.1.2.10.2."

Page 10, paragraph 3.4.1.3.2: Delete and substitute:

"3.4.1.3.2 Modulator (Speech Processing) Module - The modulator shall contain all circuits to amplify the audio signal, accomplish sideband generation and upper sideband selection in the frequency band 250 to 254 kc. It shall be capable of high level clipping and low level (base) threshold adjustment in the band 250 to 254 kc. Potentiometers shall be provided to permit adjustment of the carrier balance, the reinserted AME carrier level, audio input level, high level clipper and low level threshold adjustment. The module shall also contain circuits to control the carrier level in the AME mode in order to avoid dissipation in the RF output tubes, with or without modulation."

*Page 10, paragraph 3.4.1.3.3.2: Delete and substitute:

"3.4.1.3.3.2 Reference Oscillator Module - The reference oscillator assembly, including necessary frequency divider circuits, the SSB demodulator, and the automatic level control shall be packaged in a separate plug-in type module. The final frequency determining oscillator (reference) shall be a sealed assembly which shall be capable of maintaining stability characteristics, including aging, within 0.5 ppm for long term (30 days) and 0.2 ppm for maximum short term frequency variation (30 minutes). The SSB demodulator shall be a product type detector. The automatic level control shall be operating during AME, SSB and FSK modes."

*Page 11, paragraph 3.4.1.3.4: Delete and substitute:

"3.4.1.3.4 Sidetone Gate - BFO Module - The sidetone gating circuit and relay shall be included in the 250 kc BFO subassembly. During AME, SSB, and FSK transmission the sidetone shall be derived from the audio. Sidetone shall be gated from the rectified RF output with a suitable delay to keep the gate open during syllabic intervals. During cw transmission the sidetone shall be derived from the 400 cycle line."

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Page 22, paragraph 6.3, Precedence of Documents: Add:

"(4) Reordered Equipment - The following specifications shall take precedence over those concerning performance requirements listed or referenced in paragraph 2.1, where equipment is reordered, except in those areas where the contract specifies redesign or modification affecting performance:

MIL-E-5400D	30 November 1959
MIL-T-5422D	1 November 1957
MIL-I-6181B	29 May 1953"

*An asterisk preceding a paragraph number denotes a change in that paragraph from Amendment -2.

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